

Learning Objective 2.1: I can describe the radiation behavior of simple resonant antennas (isotropic radiator, half-wave dipole, monopole, loop) and calculate their gain and impedance.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **A whip for the guard frequency.** You are asked to build a quarter-wave monopole for $f = 121.5$ MHz on a large aluminum ground plane, fed with $50\ \Omega$ coax. Take the thin-wire half-wave dipole as $73 + j42.5\ \Omega$.

(a) How long is the whip?

$$L =$$

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(b) State the input impedance you expect at the base of the whip, and explain in one or two sentences why it is not the dipole value.

(c) Find $|\Gamma|$ and the VSWR seen by the $50\ \Omega$ feed.

$$VSWR =$$

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(d) You may change the whip or the ground plane, but you may not add parts. Give two changes that improve the match, and say what each one does to the impedance.

2. **Which way do you lay the wire?** A perfectly conducting deck acts as an image plane. An antenna at height h above it radiates as the element pattern times a two-element array factor, $2|\cos(kh \cos \theta)|$ for a vertical element and $2|\sin(kh \cos \theta)|$ for a horizontal one, with θ measured from the vertical.

- (a) A horizontal wire sits at $h = 0.05\lambda$. By how many dB is its straight-up ($\theta = 0$) field below that of the same wire at $h = 0.25\lambda$?

 $\Delta =$

- (b) Evaluate the vertical array factor at the horizon ($\theta = 90^\circ$) for an arbitrary height h . What does the result tell you?

AF =

- (c) In one or two sentences, explain why a horizontal wire lying directly on a metal deck radiates essentially nothing.
- (d) You must put up a field-expedient antenna on a large metal roof for horizon-to-horizon communication, and you can only get about 0.05λ of height. Which orientation do you choose, and why?

3. **Same radome, three antennas.** A vendor ships a vertical antenna for $f = 450$ MHz sealed in a fiberglass radome, with no datasheet. From the outside it could be a quarter-wave monopole, a sleeve dipole, or a folded dipole. You mount it on a wooden mast and run the $50\ \Omega$ coax straight down the mast.

(a) With the coax taped flat to the mast the analyzer reads a VSWR of 1.3. Re-route the coax away from the mast, or grab it with your hand, and the VSWR jumps to 2.4. Which of the three antennas is it most likely to be, and why does the cable change the reading?

(b) The same vendor sells a “ground-independent” model in an identical radome: a folded dipole with a 4:1 balun in its base. Take a resonant half-wave dipole as $73\ \Omega$, purely resistive. Find the impedance at the folded dipole’s terminals, the impedance the balun presents to the coax, and the VSWR on the $50\ \Omega$ line.

$R_{\text{fold}} =$

VSWR =

(c) Keep the first antenna and the wooden mast. Give two changes, without replacing the antenna, that make its VSWR stop depending on the cable.

4. **The full loss budget of a magnetic loop.** A single-turn circular loop of radius $a = 0.30$ m is made of copper wire of radius $b = 3$ mm and operated at $f = 14.2$ MHz. At that frequency the copper surface resistance is $R_s = 0.98$ m Ω per square. Recall $R_r = 20\pi^2 (C/\lambda)^4$ and $R_{\text{ohmic}} = (C/2\pi b) R_s$ for a single turn.

(a) Compute C/λ and confirm the loop is electrically small.

$C/\lambda =$

(b) Find the radiation resistance.

$$R_r =$$

(c) Find the ohmic resistance and the radiation efficiency. Express the efficiency in dB and give the gain in dBi.

$$\eta_{\text{rad}} =$$

$$G =$$

(d) The same loop is a respectable receiving antenna at 14.2 MHz but a poor transmitting one. Explain the difference in one or two sentences.

5. **Loop or dipole?** Continue with the loop of the previous problem.

(a) A direction-finding team rotates a small loop to take a bearing. Why do they turn the loop for a *null* rather than for a peak, and where does that null point relative to the loop?

(b) A vertical half-wave dipole and a small horizontal loop are both mounted on a mast. What polarization does each radiate toward a distant receiver on the horizon?

(c) Rewind the loop with $N = 20$ turns. Radiation resistance scales as N^2 and ohmic resistance as N . Find the new efficiency.

$\eta_{\text{rad}} =$

(d) If turns are that effective, why is a 20-turn loop still not the antenna you would choose to transmit with? Name two reasons.

6. **The price of being small.** Stay with the single-turn loop of radius $a = 0.30$ m at 14.2 MHz ($\eta_{\text{rad}} = 0.113$). The Chu limit from Lesson 3 gives the radiation quality factor of any antenna that fits inside a sphere of radius a as $Q \gtrsim 1/(ka)^3 + 1/(ka)$. Lesson 3's fractional bandwidth is roughly $1/Q$ at the half-power points, or $0.71/Q$ at the course's VSWR ≤ 2 bar.

- (a) Compute ka , the Chu Q , and the resulting VSWR ≤ 2 bandwidth in kHz.

$$Q = \boxed{}$$

$$\Delta f = \boxed{}$$

- (b) A single-sideband voice channel needs about 3 kHz. Does the loop pass it, and what does your answer imply about operating across a 350 kHz amateur band?

- (c) Measured on a VNA, this loop shows a 2:1 VSWR bandwidth several times wider than your Chu estimate. Why? Is that good news?

- (d) State the size-bandwidth trade as a rule of thumb, in one sentence.

Documentation: _____